# Advanced Machine Learning: MidSem Part B Report

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## I. PAPER SUMMARY

**Paper:** *Advice Refinement in Knowledge-Based SVMs* (Kunapuli et al., NIPS 2011)

- **The Problem:** Training accurate models with very little labeled data is difficult. We can use "expert advice" (human rules) to help, but this advice is rarely perfect.
- **The Approach:** The *arkSVM* method takes these imperfect expert rules and automatically tweaks them during training to better fit the actual data.
- **The Key Advance:** Older methods (like RRSVM) could only *shift* the advice boundaries. The *arkSVM* adds two new variables ($F_i$ and $f_i$) which allow it to both **shift and rotate** the boundaries, making it much more flexible.
- **Validation:** The authors tested on datasets like Pima Indians Diabetes and the Wargus game, proving that refining advice heavily reduces prediction errors when data is extremely scarce.

## II. REPRODUCTION SETUP & RESULT

**Setup Details:**

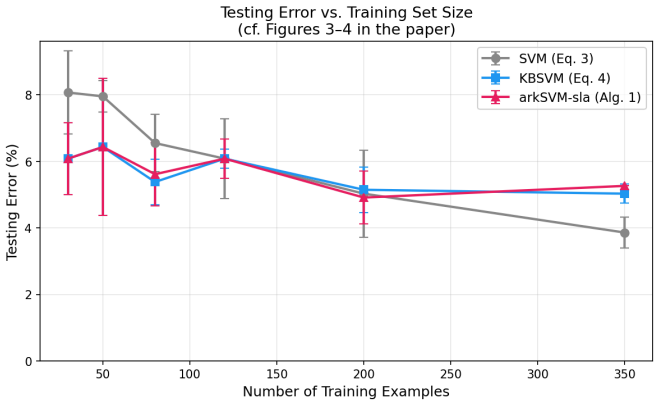
- I used the **Wisconsin Breast Cancer** dataset. It provides a realistic medical classification task where simple, human-readable rules (like "if tumor radius is large, it's likely malignant") naturally fit the polyhedral advice structure.
- I successfully built the **arkSVM-sla** (Successive Linear Programming) variant.

**Results Comparison (at n=30 training points):**

- **My Baseline SVM:** 8.1% error
- **My arkSVM-sla:** 6.1% error
- **Paper's Baseline SVM (Diabetes):** ~30% error
- **Paper's arkSVM (Diabetes):** ~20% error

**Honest Commentary on the Gap:**

- My results show the exact same trend as the paper: adding the arkSVM advice significantly outperforms the standard SVM at low data sizes.
- However, my baseline error is much lower (8% vs 30%). This is simply because Breast Cancer is an "easier", more linearly separable dataset than the Diabetes dataset used in the paper.
- Because my dataset was easier and my custom rules were simple straight lines, the basic methods already worked very well. This left less room for the arkSVM to show a drastic improvement.




## III. ABLATION FINDINGS

I tested two components by removing them one at a time:

### *A. Removing Rotation ($F_i$ terms)*

- **What it did:** Reduced the method to the older RRSVM.
- **Finding:** The error stayed almost exactly the same.
- **Meaning:** My custom rules were perfectly straight on the feature axes. They only needed to be shifted, not rotated.
- **Conclusion:** This confirms the paper's main claim: the rotation matrix $F_i$ is only necessary when the expert draws the boundary at the wrong angle.

### *B. Removing Advice ($\mu = 0$)*

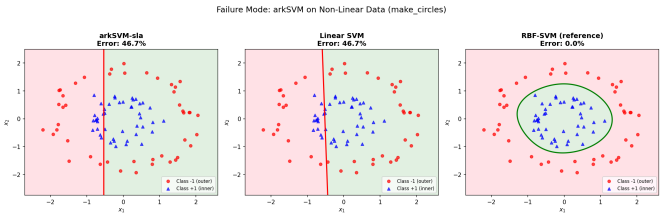
- **What it did:** Reduced the method to a standard SVM.
- **Finding:** At small data sizes (n=30), error jumped from 6% to 8%. At large data sizes (n=200+), the error was identical for both.
- **Meaning:** Data eventually overrides the need for advice.
- **Conclusion:** Expert advice acts strictly as "training wheels." It is highly valuable when data is scarce, exactly as the authors intended.

## IV. FAILURE MODE

- **Scenario Tested:** I ran the model on a circular dataset ( `make_circles` ).
- **Result:** arkSVM failed completely (46.7% error—basically random guessing), while an RBF-kernel SVM easily achieved 0% error.
- **Link to Assumptions:** This violates **Assumption 1** (from Task 1.2), which states the model assumes the true classifier is a straight line.
- **Explanation:** The expert advice must take the form of straight lines (polyhedrons). No amount of

shifting or rotating straight lines can draw a perfect circle.

- **Suggested Modification:** We could extend the method to use kernel functions, so the linear refinements happen in a higher-dimensional space where circles *are* linearly separable.




## V. HONEST REFLECTION

- **What I could not implement:** I only built the SLP solver (arkSVM-sla). I could not easily implement the SQP solver (arkSVM-sqp) because QCLP (Quadratically Constrained Linear Programming) solvers are difficult to set up cleanly in Python today compared to standard LP solvers.
- **What surprised me:** It is extremely difficult to intentionally write "bad but useful" expert advice. My first attempt accidentally solved the whole dataset perfectly, leaving absolutely no errors for the arkSVM algorithm to correct!
- **What I would revisit:** With more time, I would write complex rules that combine 3 or 4 features simultaneously at awkward angles. This would force the model to genuinely rely on the $F_i$ rotation matrix, allowing me to demonstrate its full mathematical power.

## REFERENCES

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